Effect of internal stress on the electro-optical behaviour of Al-doped ZnO transparent conductive thin films

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In this work, we will report on scientific efforts aimed at unraveling the quantitative effect of elastic strain on the electro-optical behaviour of Al-doped zinc oxide (AZO). AZO thin films have been deposited by reactive magnetron sputtering to thicknesses from 300 to 500 nm, both on extra-clear glass substrates as well as on oxidised Si wafers. This resulted in both cases in polycrystalline, strongly textured (002) films. During deposition, the internal stress evolution in the growing film was monitored *in-situ* using high resolution curvature measurements. The resulting growth-induced elastic strain, which was found to depend heavily on the oxygen partial pressure, could further be modulated by appropriately choosing the deposition temperature. The latter also induces an additional extrinsic thermal stress component, whose sign depends on the substrate used. As such, a wide range of biaxial internal stresses could be achieved, from −600 MPa in compression up to 800 MPa in tension. The resulting charge carrier mobilities, obtained independently from room temperature Hall measurements, were found to range between 5 and 25 cm^2/V s. Interestingly, the maximum mobility occurred at the zero-stress condition, and together with a charge carrier concentration of about 8×10^{20} cm^{-3}, this gave rise to a resistivity of only 300 $\mu\Omega$ cm. From the stress-dependent optical transmission spectra in the range of 200–1000 nm, the pressure coefficient of the optical bandgap was estimated from the corresponding Tauc plots to be 31 meV/GPa, indicating a very high strain-sensitivity as well. *Published by AIP Publishing*. [http://dx.doi.org/10.1063/1.4960580]

I. INTRODUCTION

Metal oxides are technologically important for use in next generation electronic devices that require transparency, conductivity, and chemical stability. In this respect, transparent conducting oxides (TCO) must be transparent like glass but also conductive like metals, although they typically have about 100 times less electrons.[1] TCOs are an important constituent of 2nd and 3rd generation solar cells. They provide electrical contact to the absorbers in order to efficiently collect the generated charges. They can also provide light trapping patterns to increase the efficiency of some cells such as those made from amorphous and microcrystalline silicon.[2] Besides their obvious relevance in solar cells, TCOs are also used in mirrors and in low emissivity glasses to avoid heat waste in the environment, and as thin film transistors in transparent displays.[3]

The majority of crystalline TCO's are made from five basic sets of metal oxides: SnO_2, In_2O_3, ZnO, CdO, and Ga_2O_3, each oxide needing further appropriate doping.[4] At present, Sn-doped In_2O_3 (ITO), Al-doped ZnO (AZO), and F-doped SnO_2 are by far the most commonly used TCO materials, since their bandgap is sufficiently large to be transparent and sufficiently small to allow efficient doping. Although ITO still tends to be considered as the reference material for most industrial applications, the development of impurity-doped ZnO and SnO_2 is becoming increasingly important. Indeed, the expanding need for transparent electrodes for optoelectronic device applications is starting to jeopardize the availability of ITO's main constituent indium, a very expensive and scarce material. In this respect, Al-doped ZnO (AZO) semiconductors consist of source materials that are inexpensive and non-toxic, and are therefore considered to be very promising alternatives to ITO for thin film transparent electrode applications.

With respect to AZO fabrication, the manufacturing of photovoltaic modules has been identified worldwide by the glass industry to be(come) a major product in their technology development roadmap. Knowing that the mass of coated glass produced by an average sized float glass melter easily exceeds 500 tons/day, the deposition rate to be achieved ultimately in large volume manufacturing of thin films must necessarily be very rapid as well, especially in the case of transparent conducting oxides which may have thicknesses on the order of 500 nm. Although many basic research issues still ought to be addressed first in laboratory R&D, the need of the eventual large-scale manufacturing must be anticipated already at the deposition stage. In this respect, fabrication methods for industrially relevant TCO films include sputtering and chemical vapour deposition (CVD). Both are well-documented to accommodate large substrate areas, and both can be designed for rapid rates of deposition.[5] Of the two methods, CVD has probably the lower capital cost and has already been demonstrated on an industrial scale to be capable of producing mechanically adhering films on large areas at acceptable electrical and optical properties.[6] On the CVD-side, (doped) SnO_2-based systems are currently emerging as the TCO material of choice for large-scale PV

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applications. In the case of sputtering, either metallic or metal oxide targets can be used. Metallic targets are cheaper to manufacture but, due to the required reactive sputtering mode needed to arrive at metal oxide thin films, conditions have to be set very carefully to avoid target poisoning, arcing, and oxygen ion bombardment. In the case of metal oxide targets, radio-frequency (RF) sputtering must be used, which is more difficult to scale up due to its inherently slower deposition rate. When it comes to replacing sputtered In_2O_3-based oxides, Al-doped ZnO can definitely be considered to be the most promising sputter-deposited TCO material for large-scale PV deployment.[7] Moreover, as compared to CVD SnO_2, sputtered ZnO-based films are also more readily amenable to chemical etching, making them also the preferred TCO material when it comes to generating rough interfaces by surface texturing for increased light trapping capabilities.[8]

Recently, we already reported on the evolution of internal stress during RF sputtering of Al-doped ZnO.[9] These experimental observations were derived from a novel *in-situ* diagnostic tool, based on high-resolution *in-situ* curvature measurements which allow to detect constrained volume changes in the course of thin film deposition. A specific asset of these high-resolution in-situ curvature measurements for monitoring the film growth process is that it enables at the same time to quantify the internal stress evolution developing in growing thin films. Indeed, curvature and stress are geometrically related in a very straightforward way in the case of thin films on elastic substrates, since both result from constrained volume changes.[10] The microstructural origin of such growth-induced internal stresses can be very diverse, and include, besides point defect generation (like vacancies and interstitials), also grain evolution (island vs. terrace growth), crystallisation, and phase transformations.[11] Such high-resolution curvature measurements have therefore been proven in the past to be a very useful *in-situ* diagnostic tool for monitoring on-line the evolving microstructure of growing thin films. In contrast to the vast amount of work on polycrystalline metallic thin films,[12,13] only few *in-situ* curvature measurements have been reported during the growth of metal oxide thin films.[14] Important fundamental differences between metallic and metal oxide thin films include the stoichiometry of metal oxide compounds, as well as differences in mobility and atomic bonding (metallic vs. ionic or covalent). The latter, for instance, can be expected to change the energy balance for crystallite coalescence[15] and therefore might reveal some new features of the internal stress evolution as well.

To the best of our knowledge, only a few other references are available in the literature showing *in-situ* measurements of the internal stress evolution during a TCO film growth process. These involve mainly systematic efforts by the group of Dr. Ellmer at the Hahn-Meitner-Institut in Berlin,[16] as well as Professor Wuttig's lab at RWTH Aachen.[17] With respect to these efforts, our own ambition in the current work has been to more actively look into the possibility of strain engineering as an appealing way to improve the functional properties of TCO layers. Indeed, most of the current state-of-the art research looks at internal stress essentially as a detrimental mechanical durability issue, with issues like delamination or fracturing of the oxide films. In the current work, we aim to look more actively at internal stress and the associated elastic strain as a potentially useful instrument for increasing the functionality of TCOs. In this respect, it has already been reported in the literature, for instance, that DC sputtered Al-doped ZnO films with the lowest resistivity generally also had the lowest level of structural strain,[18,19] the latter being determined from XRD diffractograms on as-deposited samples. Other reports have suggested a direct link between the crystal quality of RF sputtered ZnO in terms of film texture and stoichiometry, and the level of internal compressive stress derived *ex-situ* from peak shifts in micro-Raman spectra.[20] While the majority of the characterisations reported in the literature on the internal stress level in TCO films are based on *ex-situ* measurements in as-deposited samples, it is clear that efforts to further unravel the link between the internal stress and the resulting electro-optical and structural properties would benefit largely from *in-situ* measurements of the internal stress evolution taken in real-time during the deposition process itself.

With respect to its electro-optical behaviour, any TCO material must necessarily represent a compromise between the conflicting properties of electrical conductivity and optical transmittance. According to Eq. (1)

$$\frac{1}{\rho} = e \cdot \mu_H \cdot N_d, \tag{1}$$

a reduction in resistivity ρ involves an increase in either the free carrier concentration N_d or in the Hall mobility μ_H. While increasing the carrier concentration generally also leads to a shift in plasma frequency, increasing the mobility has usually no detrimental effect and is therefore considered to be the preferred research direction to follow. In this respect, for the specific framework of photovoltaics, research goals have been set to aim for a resistivity of less than 400 $\mu\Omega$ cm, which would correspond to a targeted sheet resistance of less than 10 Ω/sq for a TCO film with a typical thickness of 500 nm. Setting targets on the optical properties should be done with more care, as the reflectance and/or transmittance of TCO films exhibits interference fringes in the operational spectrum of PV devices. Although the transmittance may peak for specific wavelengths at values as high as 85%–90%, defining an absorption coefficient or an optical band gap is physically more useful. For a 500 nm thick film, besides a target resistivity of 400 $\mu\Omega$·cm, an absorption coefficient due to free carriers of $2 \times 10^3\,cm^{-1}$ for a material with a bandgap of at least 3 eV can be considered a difficult but feasible goal.[21] In order to reach these properties, this comes down to achieving free carrier mobilities on the order of 50 cm^2/V s.

As a summary, Figure 1 provides the bigger picture of ZnO's electronic behaviour, showing Ellmer's compilation[22] with state-of-the-art data for defect-free, mono-crystalline bulk ZnO (both undoped and In-doped) of RT mobility μ_H as a function of carrier density N_d. We have slightly extended this graph with the iso-resistivity contour of 100 $\mu\Omega$ cm,

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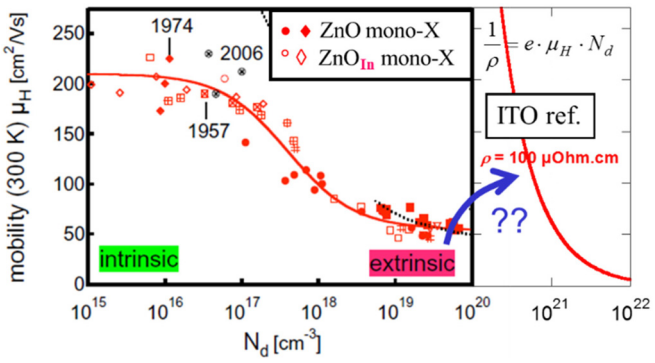


FIG. 1. Ellmer's compilation[22] of RT mobility data μ_H as a function of carrier density N_d for defect-free, mono-crystalline bulk ZnO, both undoped and In-doped. We have extended this graph with the iso-resistivity contour of 100 $\mu\Omega$ cm associated with the benchmark ITO system.

illustrating the remaining gap which still exists between ZnO-based systems and the electrical behaviour of the benchmark ITO system.

II. EXPERIMENTAL

As to the AZO thin film deposition, a turbo-pumped AJA deposition chamber with five 2 in. magnetron guns was used in sputter-up configuration. During reactive sputtering, the Ar and O_2 flows were regulated by independent mass flow controllers, while the total sputtering pressure, as measured by a capacitance vacuum meter (Baratron), was kept constant at 2 mTorr. In our deposition chamber, which was already depicted and described in detail in a previous publication,[9] the problematic bombardment of negative oxygen ions has been minimized, thanks to the relatively large target to substrate distance (122 mm), combined with a rotation of the substrate (at 40 rpm) and the delivery of the oxygen gas through small holes disposed along a ring directly at the level of the substrate.

In order to arrive at sufficiently high carrier concentrations, our ZnO films have been doped extrinsically with aluminum during the deposition process, based on a reactive sputtering process from an alloyed metallic Zn(1.3 wt. % Al) target. Films were deposited both onto 3 in. wafers and (1 × 1 inch) squares cut from 380 μm thick Si wafers which had been thermally oxidized by a LPCVD wet oxidation process at 1000 °C to grow a SiO_2 layer of 540 nm. For the optical measurements, AZO films have also been deposited directly onto 1 × 1 inch squares, 400 μm thick extra clear selected float glasses (low iron content). After transfer into the ultra-high vacuum chamber, the substrates were first precleaned with an RF-bias for a few minutes during which Ar atoms accelerated at 150 eV to remove about 3–5 nm of the substrate.

As to the internal stress measurements, this can be done in a variety of ways, such as from the shift of XRD peaks[23] or from the bending of a cantilever or an unconstrained wafer.[24,25] The latter technique, which is usually based on the so-called Multi-beam Optical Stress Sensor (MOSS), has already proven to be very efficient for *in-situ* internal stress monitoring in the case of metal evaporation[24] or CVD growth of epitaxial films.[25] We have used the same MOSS-technique (from k-Space) on our sputtering chamber to derive the internal stress evolution in real time. The combination of parallel beam illumination and CCD detection results in a system that is capable of measuring a radius of curvature as large as 10 km, with a time resolution on the order of 0.1 s. In order to suppress the light from the plasma discharge, a band pass filter (660 nm ± 4 nm) was mounted in front of the CCD. During deposition, a change in curvature $\Delta\kappa$ is thus measured with a laser deflectometer as shown in Figure 2, which illuminates the sample with an array of 12 parallel laser beams, whose reflected position and intensity is then continuously acquired by a CCD camera. Since the substrate was rotated for better deposit uniformity, the CCD was triggered with a proximity sensor at every rotation. A framegrabber then records in real time the centroid position of each spot, i.e., the equivalent of their gravity center. According to Eq. (2)

$$\Delta\kappa = \frac{D - D_o^*}{D_o^*} \cdot G, \tag{2}$$

the curvature change in the two in-plane directions of the deposited film is simply given by the mean differential spacing (D_o^* being the spot spacing when the substrate is replaced by a perfectly flat mirror) times the optical distance G (0.68 m in our case, as determined with a set of calibrated curved mirrors). The curvature change measured along the two in-plane directions of the film is then directly proportional to the product of the mean internal stress $\langle\sigma_f\rangle$ and the film thickness h_f according to the following well-known Stoney equation:

$$\frac{Y_s \cdot h_s^2}{6} \cdot \Delta\kappa = \langle\sigma_f\rangle \cdot h_f = \int_0^{h_f} \sigma_{incr} \cdot \mathrm{d}h_f, \tag{3}$$

where Y_s is the biaxial modulus of the substrate and h_s is the substrate thickness. The substrate biaxial modulus may be expressed in terms of Young's modulus E and the Poisson coefficient ν as $Y_s = E/(1-\nu)$, and amounts to 182 GPa or 92 GPa for a (111) silicon or glass substrate, respectively. Since the deposited film is much thinner than the substrate, one advantage of this technique is that there is no need to know the modulus of the deposited film itself. In that case,

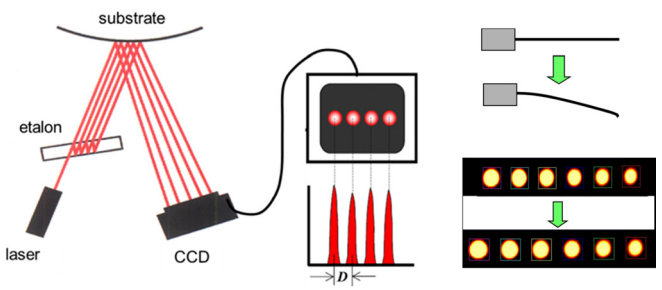


FIG. 2. Working principle of the so-called Multi-beam Optical Stress Sensor (MOSS). During deposition, a change in curvature is measured with a laser deflectometer, which illuminates the sample with an array of parallel laser beams. The reflected positions and intensities are then continuously acquired by a CCD camera, in which a framegrabber records in real time the centroid position of each spot.

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the film might even be subject to plastic deformation without any consequence for the applicability of Stoney's film stress-substrate curvature relation. Possible variations in deposition rate were measured *in-situ* by a quartz crystal microbalance, and used, together with the final film thickness measured *ex-situ* with cross-sectional SEM, to convert the time scale during deposition into a thickness scale. As such, the measured stress*thickness product can then be plotted as a function of the film thickness. In such a graph, according to Eq. (3), the slope of a line connecting the origin to each point of such a curve represents the evolution of the film mean internal stress $\langle\sigma_f\rangle$ during growth, i.e., the internal stress averaged through the film thickness at any given point in time. Moreover, the local slope can also be derived to obtain the evolution of the so-called instantaneous or incremental internal stress σ_{incr}, associated both with growth of new film, and relaxation occurring within the already deposited film. As illustrated schematically in Figure 3, positive values of the stress*thickness product imply a tensile mean stress, while negative values imply a compressive mean stress. Similarly, positive (negative) slopes correspond to a tensile (compressive) instantaneous stress.

Samples were also characterized *ex-situ* after deposition. The film thickness was checked by cross-sectional SEM (Zeiss Ultra 55 field-emission-gun) after each deposition. The phase structure was determined by X-ray diffraction, measured with Cu $K\alpha$ radiation in the usual θ-2θ geometry. Sample composition (purity and Al-concentration) was checked by ICP/OES measurements (Varian 720ES) on layers dissolved in an acid mixture. The sheet resistance at room temperature was determined by a four-point probe, while a room temperature Hall measurement set-up (HMS-3000 from Microworld) was used in van der Pauw configuration to independently determine mobility and charge carrier concentrations. Finally, optical transmission spectra were obtained with a spectrophotometer (Specord 200 PLUS from Analytic Jena) in the range of 200 to 1100 nm with a 0.5 nm resolution.

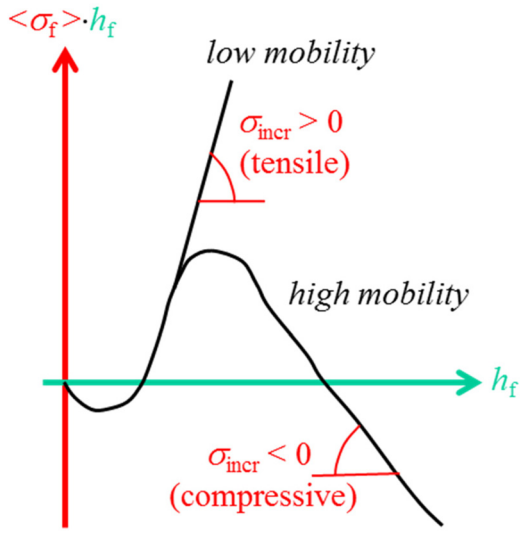


FIG. 3. Typical evolution of the stress*thickness product as a function of film thickness. In such a graph, according to Eq. (3), the slope of a line connecting the origin to each point represents the evolution of the mean internal stress $\langle\sigma_f\rangle$. Moreover, the local slope at each point describes the evolution of the so-called instantaneous or incremental internal stress σ_{incr}. Positive values of the stress*thickness product imply a tensile mean stress, while negative values imply a compressive mean stress.

III. RESULTS AND DISCUSSION

A. Internal stress evolution

In a first attempt to optimise the processing window for our reactively sputtered Al-doped ZnO thin films, a discharge curve was established at room temperature as a function of oxygen partial pressure (p_{O_2}) in the sputtering chamber. The result, shown in Figure 4 for a total chamber pressure of 2 mTorr, shows 2 characteristic sputtering modes, i.e., a metallic one at low p_{O_2} and an oxide one at high p_{O_2} values, respectively. Although for process and plasma stability reasons, it is often suggested to stay in either of both working modes, we have chosen in this work to rather fix our p_{O_2} to be in the transition regime at 20% O_2. As will be shown further on, this has allowed, in combination with a substrate temperature variation, to generate the intended wide experimental range of charge carrier concentrations.

With respect to the deposition temperature, Figure 5 shows its effect on both the deposition rate (in Å/s) and the Al-content (in wt. %) over the considered range from room temperature (RT, no intentional heating) to 250 °C. From this graph, it is clear that a temperature of 250 °C can be considered to be already relatively high in the case of reactive sputtering from an alloyed Zn-target. Indeed, Zn being a relatively volatile metal ($T_m \cong 420$ °C), its equivalent homologous temperature (T/T_m) amounts to almost 0.8. As a result, a sharp decrease in deposition rate (factor 3, from 1.2 to 0.4 Å/s) is observed from 200 °C onwards, which can be attributed to a significant amount of Zn evaporation. The latter was also found to be accompanied by an increase in Al-content, from about 1.2 wt. % at RT to more than 2.5 wt. % at 250 °C. Based on these rather elementary observations, it is clear that the films deposited at 250 °C should be considered in the forthcoming discussions to be chemically very

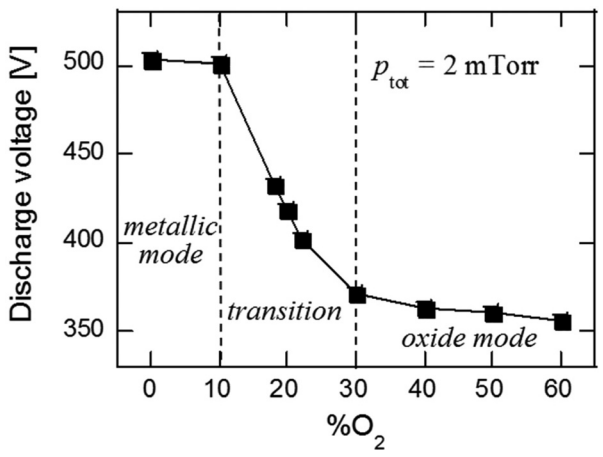


FIG. 4. Typical discharge curve of DC target voltage as a function of oxygen content in the sputtering chamber for a total pressure of 2 mTorr. In this work, the oxygen content during reactive sputtering was fixed to be in the transition regime at 20% O_2.

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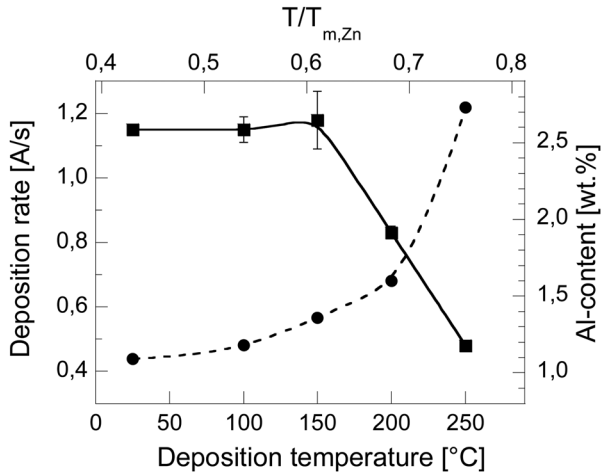


FIG. 5. Effect of deposition temperature on both the deposition rate (left, in Å/s) and the Al-content (right, in wt. %) over the considered range from room temperature (RT, no intentional heating) to 250 °C. The top abscise expresses the deposition temperature as an equivalent homologous temperature $T/T_{m,Zn}$, with $T_{m,Zn} \cong 420$ °C.

different from the ones deposited at lower temperatures. Structurally, however, all films turned out to be very similar, with a pronounced (002)-texture typical for sputtered wurtzite crystals, and an average columnar nanocrystalline grain size of about 30 nm (cf. Figure 6). The latter rather small value can be attributed to the fairly high melting point of ZnO ($T_m \cong 1975$ °C), reducing the homologous deposition temperature to a mere 0.23 once the films were condensed onto the substrate as a metallic oxide.

Figure 7(a) then shows the evolution of the internal stress as a function of thickness, as measured in real time during reactive sputtering at different substrate temperatures in the range RT-250 °C. First of all, from this graph, the specific curvature resolution obtained in the current work for reactive ZnO deposition on Si substrates can be obtained. Its value in terms of changes in radius of curvature was on the order of 5.3 km, and was calculated as twice the standard deviation of the baseground curvature signals before any film growth started (so from the data to the left of the vertical dotted line at zero thickness). Three more major observations can then be made. First of all, from the more or less constant slopes of the stress*thickness vs. thickness curves, the incremental internal stress can be taken as constant throughout the film thickness and hence, according to Eq. (3), equal to $\langle \sigma_f \rangle$. This can indeed be expected for a columnar growth process. Note that this will also further allow, in the discussion of the electro-optical properties, to exclude any effect of strain gradients within the film. Second, increasing the deposition temperature increases the compressive stress component, in agreement with the well-established underlying effect of (ad-)atom mobility on the internal stress evolution, already depicted schematically in Fig. 3. This has been associated microscopically with the incorporation of such sufficiently mobile adatoms in the grain boundary region, the latter initially being stressed in tension during grain coalescence, conveniently called "grain boundary zipping."[24] And finally, when considering quantitatively the resulting constant average stress values as a function of deposition temperature in Figure 7(b), a relatively wide range of stressed films has been covered, going from 800 MPa in tension at RT to −600 MPa in compression at 250 °C. Interestingly, as a result of the tensile-to-compressive stress transition, we have also been able to generate stress-free films, which can be taken as a very useful reference when discussing any potential stress-effect. It should be noted that 2 different average stress-data are presented in Fig. 7(b): the so-called intrinsic, growth-induced ones, resulting only from the deposition process (open squares), and the final, internal stress values obtained after cooling down the films to RT (filled circles). The difference between these 2 series of stress data is directly related to an additional, so-called extrinsic thermal stress component in the as-grown films. When further discussing the effect of stress on the electro-optical behaviour, it is obviously the latter series that need to be considered, since all electro-optical measurements have been carried out at RT as well.

B. Effect of internal stress on electronic properties

The different films shown-above, exhibiting different internal stress values, were then characterized electrically. Results for both the charge carrier concentration N_d and the room temperature Hall mobility μ_H are shown in Figures 8(a) and 8(b), respectively. First of all, the simple fact that the carrier densities (in 10^{20} 1/cm^3) can be represented in the form of an Arrhenius-plot already indicates that they are mainly temperature-controlled, rather than being stress-controlled. Obviously, the fact that the abscise in Fig. 8(a) refers to the deposition temperature would imply that carrier activation takes place during the sputtering process itself, the resulting carrier density than being quenched-in during the relatively fast cooling down to room temperature needed for unloading the sample. Besides the linear fit itself which qualitatively confirms a thermally activated process, also the quantitative value obtained for the underlying activation energy $E_a \cong 0.084$ eV seems to be in the right order of magnitude when compared to the few available literature data for other extrinsic dopants in ZnO. For instance, Kamioka *et al.* have reported a value of 0.102 eV for Ge-doped wurtzite ZnO.[26] Since the origin of the carrier concentration in the

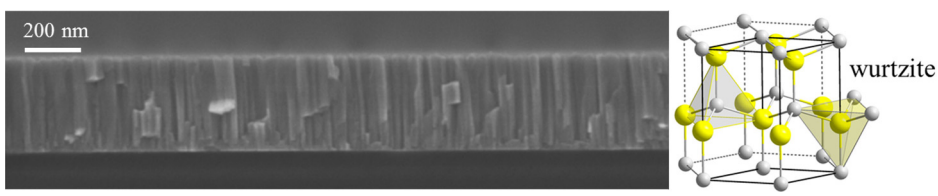


FIG. 6. Cross-sectional SEM micrograph of an Al-doped ZnO film deposited at 150 °C. All films exhibited a pronounced (002)-texture, typical for sputtered wurtzite crystals, and an average columnar nanocrystalline grain size of about 30 nm.

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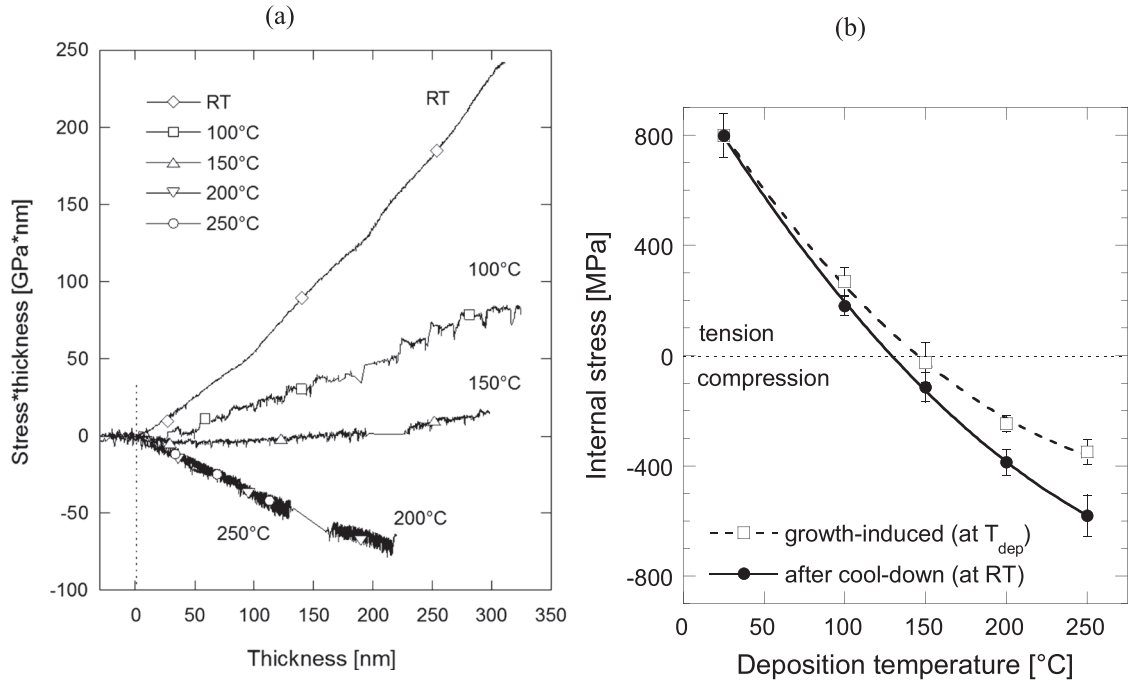


FIG. 7. (a) Evolution of the stress*thickness product as a function of film thickness, as measured in real time during reactive sputtering at different glass substrate temperatures in the range RT-250 °C. The few discrete open symbols have simply been added to be able to identify the different data series in the legend; (b) Resulting average stress values as a function of deposition temperature. Two different stress-data are presented: the intrinsic, growth-induced ones, resulting from the deposition process (open squares), and the final, internal stress values obtained after cooling down to RT (filled circles). The error bars on the data at T_{dep} represent twice the standard deviation on the slope of the linear fit of the stress*thickness data in Fig. 7(a). As no such fit is possible after cool-down, we have kept the same relative error on the RT data.

latter was shown to come from the generation of defects into the ZnO host lattice and not from Ge acting as a donor, we can also speculate that the origin of the carrier concentration in our own ZnO:Al films would come, as in the case of the cited ZnO:Ge films, from the generation of defects into the ZnO host lattice and not from Al acting as a donor. This is consistent as well with the similar values obtained in Fig. 5 for the amount of alloyed aluminium in the different films (again, except the one deposited at 250 °C). Moreover, in case defect generation in the ZnO host lattice is the main carrier activation process, it comes as no surprise that the underlying activation energy is independent of the dopant used (Al vs. Ge).

The RT mobility values in Fig. 8(b) show a much more peculiar behaviour as a function of deposition temperature. This becomes especially evident when superposing in the

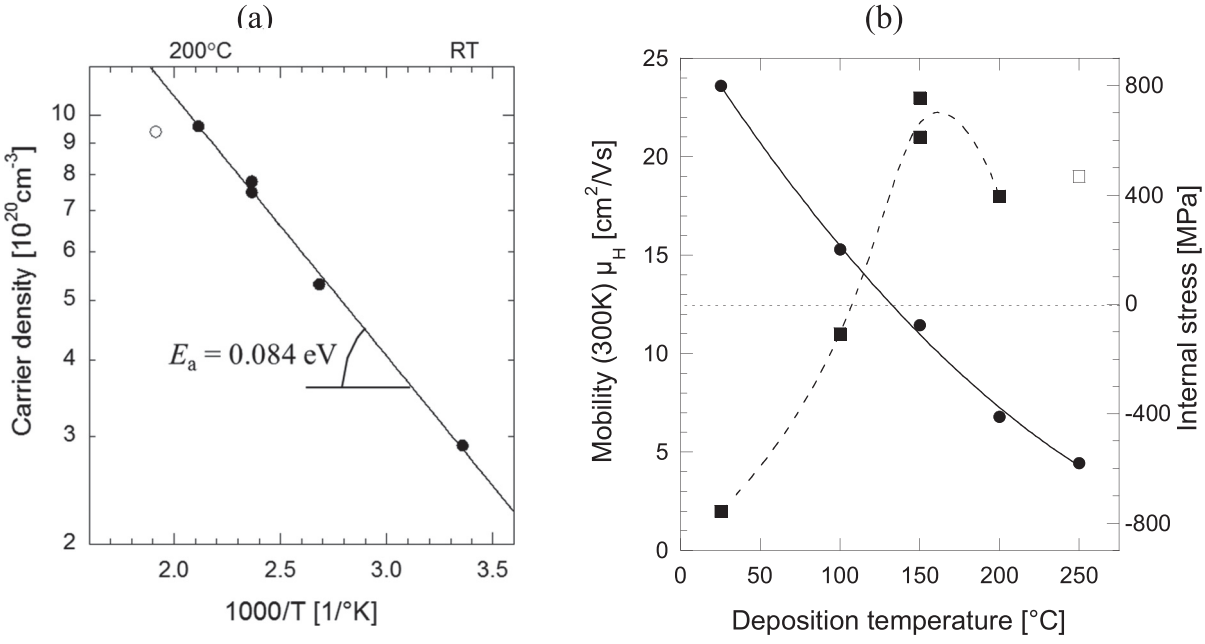


FIG. 8. Results for both the charge carrier concentration N_d (a) and the room temperature Hall mobility μ_H (b) as a function of deposition temperature.

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same graph also the temperature-dependent internal stress values of Fig. 7(b). When comparing both data sets, it becomes evident that the highest mobility is obtained for the stress-free films. In addition, from the rather symmetric mobility curve around this maximum value, it appears that only the magnitude of the stress and not its sign matters. Both observations clearly point towards a significant stress-effect governing the charge carrier mobility. Moreover, combining the maximum mobility of 25 cm^2/V s occurring at the zero-stress condition with the corresponding charge carrier concentration of about 8×10^{20} cm^{-3} then results in a resistivity of only 300 $\mu\Omega$·cm.

Note that in both Figs. 8(a) and 8(b), the data points at 250 °C represented as an open circle (Fig. 8(a)) or an open square (Fig. 8(b)) lie systematically outside the observed trends. As already argumented before, this can be attributed to a different film chemistry, namely, to a significantly higher Al-content as a result of selective Zn evaporation.

One may wonder at this stage whether the claimed stress-effect on mobility shown in Fig. 8(b) may not be attributed in part to a variation in microstructure, resulting from depositing at different temperatures. Although we already suggested earlier that, in view of the high homologous temperature of ZnO, such a variation should be minimal within the temperature range considered here, a specific experiment was designed to be more conclusive in this respect. As illustrated in Figure 9, it consisted in depositing the same Al-doped ZnO film under exactly the same conditions of T and p_{O_2} but on 2 different substrates: oxidised Si on the one hand and glass (so SiO_2 as well) on the other hand. This generates the most pure experimental condition for comparing differently stressed films at a constant microstructure. Reliable evidence for a constant microstructure is the fact that in Fig. 9, the internal stress evolution during growth was found to be independent of the substrate used, at least as long as its starting top surface was amorphous (as is the case when comparing oxidized Si to glass). The different value of the substrate's thermal expansion coefficient relative to the AZO thin film then results in the case of the oxidised Si-substrate in a further increase of the internal stress in the tensile direction during cool-down from 150 °C to RT, while the film on the glass substrate takes on the almost stress-free condition already depicted in Fig. 8(b). A clear effect on the film's resistivity is observed, with almost a factor 2 increase (from 303 to 546 $\mu\Omega$·cm) for the film stressed in tension at 202 MPa. Based on such thermal stress values measured as a function of temperature during cool-down, it was possible to back-out the elastic modulus Y_{AZO} of these films.[14] The obtained value of $Y_{AZO} = 121$ GPa then also allows to estimate the elastic strain range associated with the internal stress evolutions reported so far, strain being the more fundamental parameter governing mobility. For instance, the 202 MPa film in Figure 9 can be considered to be strained elastically to $\varepsilon_f \equiv \langle\sigma_f\rangle/Y_f = 0.17\%$.

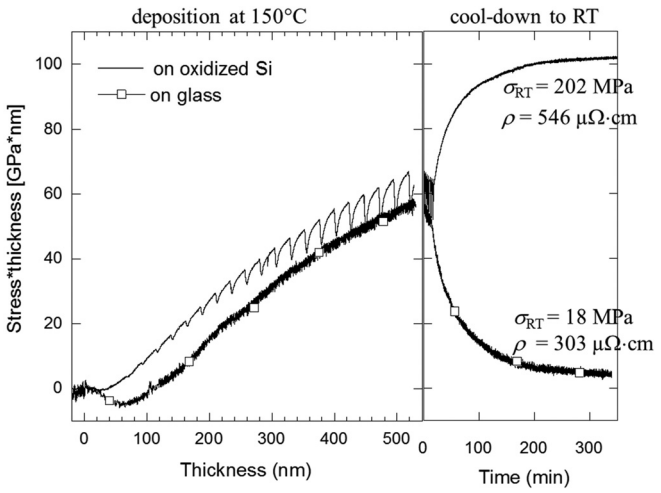


FIG. 9. Evolution of the stress*thickness product during deposition and cool-down to room-temperature under exactly the same conditions of T (150 °C) and p_{O2} but on 2 different substrates: oxidised Si on the one hand and glass (open squares) on the other hand. The periodical fluctuations of the stress*thickness data measured on oxidised Si are a result of additional optical interferences of the MOSS laser spots within the transparent SiO_2 film. The few discrete open symbols have been added "manually" to be able to identify the different data series in the legend.

A final issue then relates to the effect of the Al-doping, in other words: would the same stress-effect be observed for pure, undoped ZnO as well? We therefore repeated the same experiments for reactively sputtered pure ZnO (using a pure Zn target) in the same deposition temperature range. The resulting RT internal stress data are presented in Figure 10(a), showing a very similar qualitative behaviour as compared to the Al-doped ZnO films, including a tensile-to-compressive transition. Quantitatively, however, the internal stress values for pure ZnO were found to be significantly lower, covering the range from 200 to −400 MPa, as compared to 800 to −600 MPa for Al-doped ZnO. This can be attributed to a higher degree of strain-hardening available when Al is being included in the ZnO film. Al being a lighter element than Zn and located in a substitutional position, it is not very likely it would contribute to solid solution hardening. Therefore, it is probably the presence of a separate second Al_2O_3 phase within the ZnO matrix that is responsible for the observed mechanical behaviour. Its presence was also confirmed on the X-ray diffractograms.

The corresponding electrical behaviour is shown in Figure 10(b), representing the film's resistivity (in $\mu\Omega$·cm on the left ordinate, or in 10^{-3} Ω·cm on the right one) as a function of deposition temperature. Resistivity measurements were repeated 3 times on each sample, but the thus obtained standard deviation was smaller than the "thickness" of the data symbols used in Fig. 10(b) and hence not visible. Besides the expected significant increase in resistivity for the undoped ZnO films as compared to the Al-doped ones, which can be fully attributed to the absence of extrinsic dopants, one striking feature remains essentially unchanged when comparing to the AZO results of Fig. 8(b): a minimum in resistivity is observed in both cases at the zero-stress condition. In this respect, it is very reassuring as well to see how the shift of this zero-condition to lower temperatures in the case of undoped ZnO (essentially as a result of the absence of Al-induced strain hardening, cf. Fig. 10(a)) correlates directly with a shift to lower temperatures of the minimum in the resistivity curve. Note that, although independent values for RT mobilities and carrier densities were not

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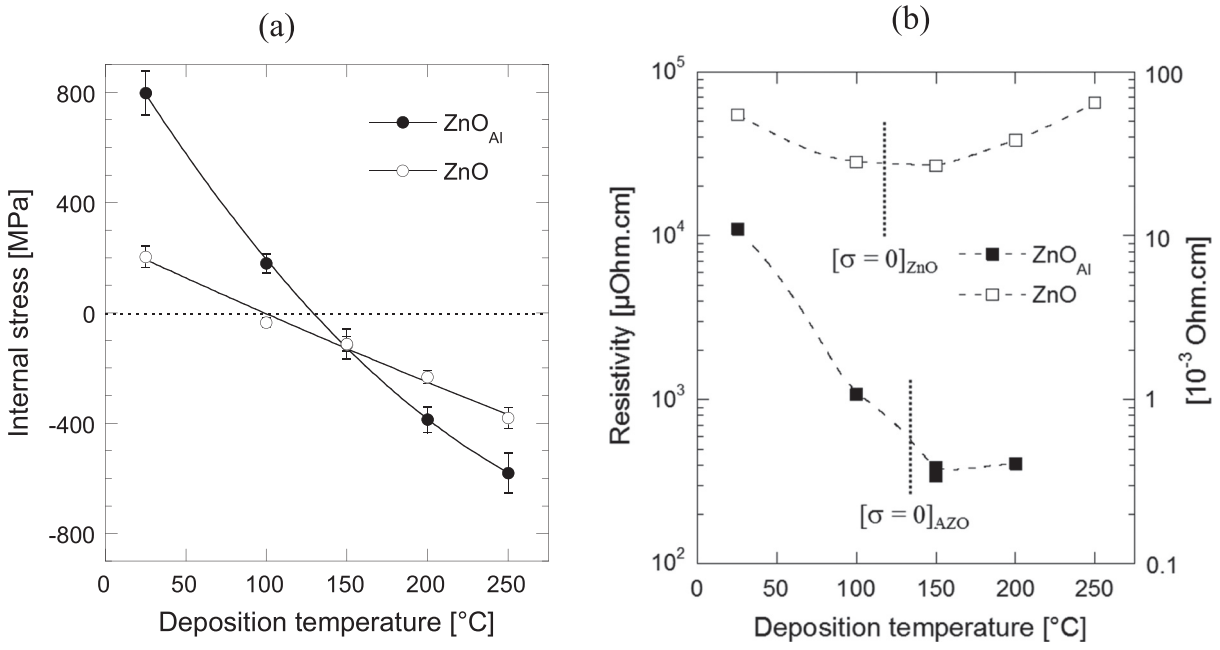


FIG. 10. (a) RT internal stress data and (b) resistivity values (in $\mu\Omega$ cm and $10^{-3}\,\Omega$ cm on the left and right ordinate, respectively) as a function of deposition temperature, comparing reactively sputtered undoped, pure ZnO to Al-doped ZnO films.

measured in the case of undoped ZnO, in view of the expected Arrhenius dependence of the latter, the observed minimum in resistivity for undoped ZnO is expected to correspond to a maximum in mobility, similarly as observed explicitly for Al-doped ZnO. Therefore, the observed significant effect of internal stress (or more correctly internal elastic strain) on the electronic properties seems to be related to the underlying ZnO wurtzite structure, which is well-known to exhibit piezo-electric properties. Such a piezo-electric behaviour can be expected to be especially pronounced in the case of our strongly (002)-textured ZnO thin films.

When comparing the above findings to results for (doped) ZnO films from the literature, we recognise that a minimum in resistivity has already been associated with strain free crystals in the past, although not with the same degree of direct evidence as we reported in the current work. For instance, Hinze and Ellmer[16] already established such a correlation in the case of RF sputtered Al-doped ZnO, but all their films were shown to be in compression, so no zero-stress condition was included. In their case, a minimum resistivity of 2000 $\mu\Omega$·cm was observed at -500 MPa. Also in the case of RF sputtered Ga-doped ZnO, the same correlation was established, with a minimum resistivity value of 2000 $\mu\Omega$·cm reported now at -800 MPa.[27] However, these films showed a maximum in both mobility and carrier density, suggesting that other, structural parameters probably interfered with the stress effect. Specifically for reactively DC sputtered AZO, Ellmer has compiled in his seminal review paper[28] a number of correlations between film stress (again only compressive values) and electronic properties. Based on his compilation, it was concluded that "films of lowest resistivity are obtained in a relatively narrow process window where large strain-free crystallites exist." Moreover, he also suggested a limiting resistivity of 200 $\mu\Omega$·cm for polycrystalline AZO, in line with the resistivity values reported in the current work. It should be noted however that from Ellmer's data compilation, it is clear that grain size is an additional critical parameter as well, which might interplay directly with the internal stress. For instance, Ellmer's conclusion about the stress-effect seems only valid at sufficiently large grain sizes.

In summary, referring back to the overall mobility vs. carrier density correlation provided in the introduction, it is clear from Figure 11 that we have been able in the current work to further approach the benchmark ITO resistivity line of 100 $\mu\Omega$·cm. This was mainly achieved however by further pushing the charge carrier concentration into de $10^{21}\,\mathrm{cm}^{-3}$ range through thermal activation, rather than by increasing the mobility into the $50\,\mathrm{cm}^2/\mathrm{V\,s}$ range. In other words, since maximum mobilities are expected at the zero-stress

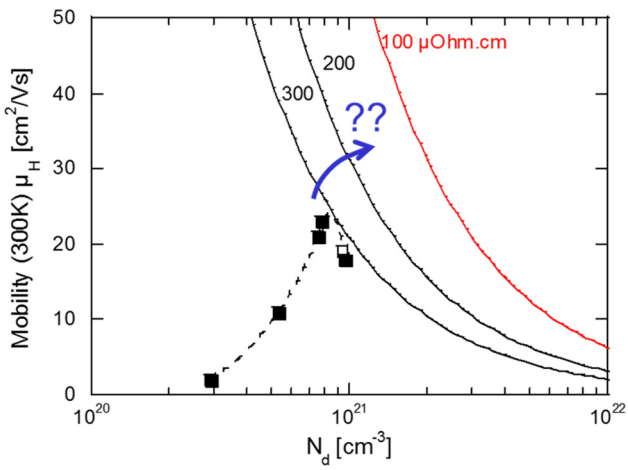


FIG. 11. Summary of the RT mobility data μ_H as a function of carrier density N_d obtained in this work on Al-doped ZnO films at various deposition temperatures. For the stress-free films, we have been able to reach the iso-resistivity contour of 300 $\mu\Omega$ cm.

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condition in the case of AZO, "strain engineering" does not really allow for much more improvement when trying to approach the resistivity of the benchmark ITO system. It is therefore expected that further optimisation of such stress-free, nanocrystalline and strongly textured films can still be achieved through "defect engineering," especially at the grain boundaries.

C. Effect of internal stress on optical properties

With respect to the optical properties, Figure 12(a) shows the transmittance spectra of the different samples as a function of deposition temperature (and hence internal stress). When integrating over the entire wavelength range from 200 to 1100 nm and normalising by the substrate transmittance, all AZO films can be considered to be sufficiently transparent (≥90%), except the RT one (cf. Fig. 12(b)). More interestingly, however, are the specific spectral features that have been unraveled for the different samples.

In this respect, we can first of all observe a blue shift of the wavelength below which all the incident light is being absorbed (zero transmittance) when increasing the deposition temperature. In view of the underlying correlation between deposition temperature and charge carrier density revealed earlier in Fig. 8(a), this effect may at first glance be attributed to the well-known Burstein-Moss (BM) effect.[29] Indeed, assuming that the conduction and valence bands are parabolic, this BM shift ΔE_g can be expressed as follows:

$$\Delta E_g = E_g - E_g^{\circ} = \frac{h^2}{8m^*} \cdot \left(\frac{3 \cdot N_d}{\pi}\right)^{\frac{2}{3}}, \tag{4}$$

where h is Planck's constant, and E_g° and m^* represent the intrinsic band gap and the free electron's effective mass, respectively. In order to quantitatively verify the above relationship with our own experimental data, values for the optical bandgap E_g were estimated first by transforming the basic transmittance spectra of Fig. 12(a) into the so-called Tauc plot of Figure 12(c). Extrapolating the linear portion of the square of the absorption coefficient α against photon energy $h\nu$ then allows to obtain the optical band gap E_g according to Tauc's proportionality[30]

$$\alpha \cdot E \sim (h\nu - E_g)^{\frac{1}{2}}. \tag{5}$$

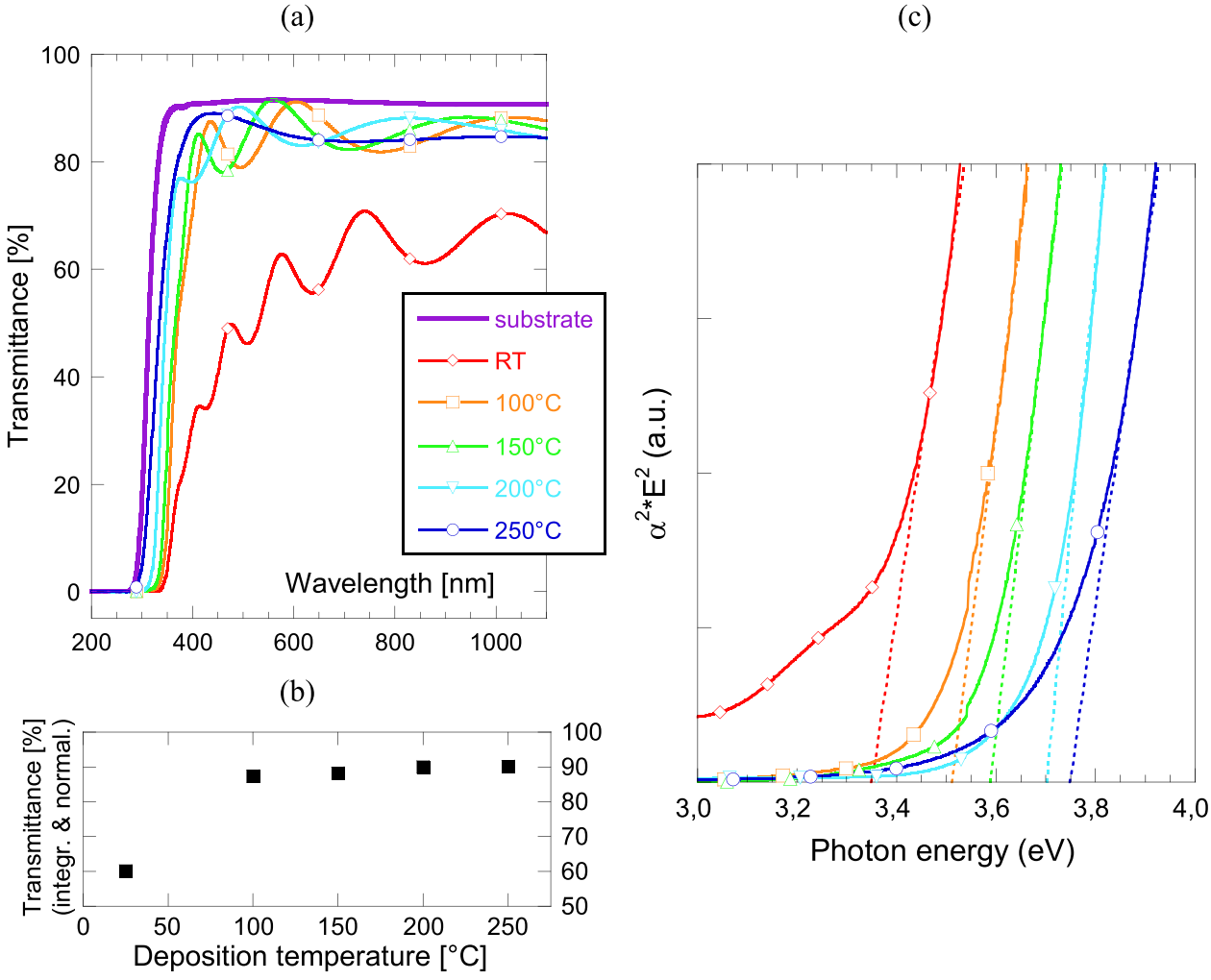


FIG. 12. (a) Transmittance spectra of the different samples as a function of deposition temperature. The few discrete open symbols have been added "manually" to be able to identify the different data series in the legend (b) Integrated spectra over the entire wavelength range from 200 to 1100 nm and normalising by the substrate transmittance. (c) Tauc plot calculated from the transmittance spectra in (a), allowing to derive E_g values by extrapolating the linear portion of the square of the absorption coefficient against photon energy.

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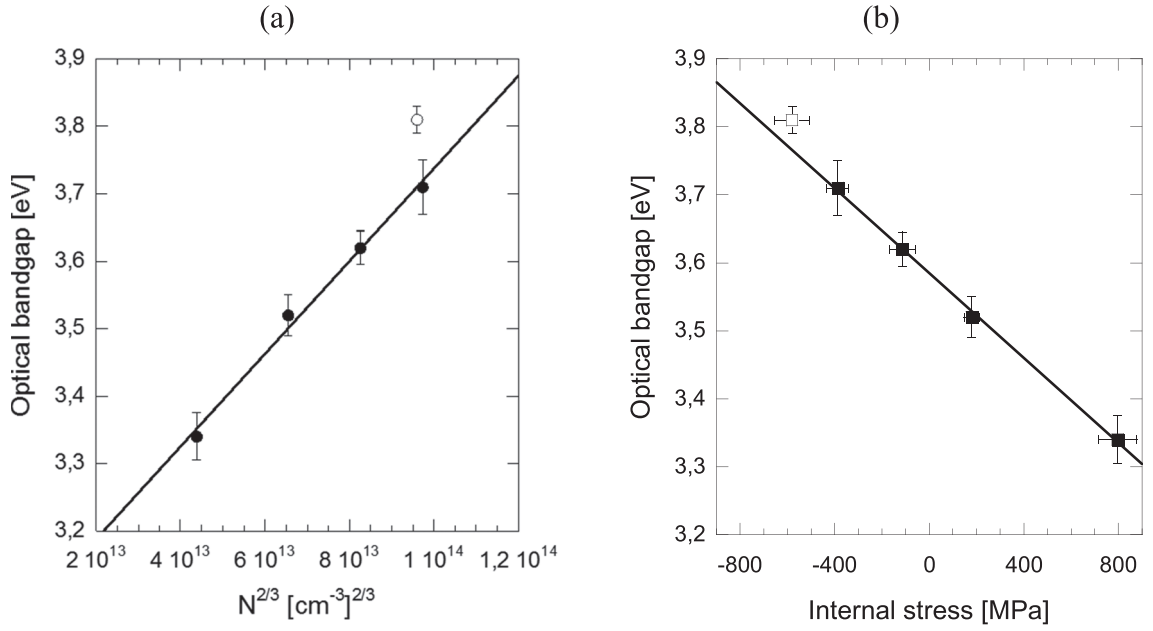


FIG. 13. Values for the optical bandgap E_g as a function of (a) 2/3th of the carrier density N_d (cf. Eq. (4)) and (b) internal stress. The error bars shown on the E_g data were calculated as twice the standard deviation on the intercept of the linear fits in the Tauc plot of Fig. 12(c).

Results are shown in Figure 13(a), which also confirms the linear relationship between optical bandgap and $(N_d)^{2/3}$ predicted by Eq. (4). Also notice that similarly as for the electrical properties, the sample deposited at 250 °C (open circle in Fig. 13(a)) can again be considered as an out-lyer. This could already be anticipated from the slopes of the linear extrapolations in Fig. 12(c), the one for the 250 °C sample being clearly different from the slope of all the other ones. Moreover, the fact that for the latter samples a single slope value was obtained is as such also fully self-consistent with the single linear fit obtained in Fig. 13(a). From this fit, Eq. (4) then finally allows to calculate a single value for the effective mass m^*, which amounts in our case to $0.54 \cdot m_e$ (m_e being the free electron mass).

This value is in reasonable agreement with other m^*/m_e values reported in the literature for Al-doped ZnO: 0.49 in Ref. 31, 0.5 in Ref. 32, and 0.6 in Ref. 33. However, all these values are about three times higher than values calculated for non-degenerate bulk ZnO.[34,35] Since the (inverse) effective mass is directly related to the band curvature, this striking deviation has been attributed by Jia *et al.*[32] to the non-parabolic nature of the conduction band, referring to the earlier work by Sans *et al.* on heavily doped ZnO:M_{III} semiconductors, with M = Al, Ga, and In.[36] In that case, however, the electron's effective mass m^* should be expected to increase even further with carrier density.[37] As this is in contradiction with the constant slope and hence constant m^* values we have observed on our AZO thin films samples in Fig. 13(a), our data might rather point to a more direct effect of strain on the optical band gap, as suggested in Figure 13(b). Indeed, rather than considering E_g to be a function only of carrier density, we believe that internal stress might be an appropriate underlying parameter as well. In that case, the pressure coefficient for the optical bandgap we obtained for our stressed AZO thin films, despite possible uncertainties inherent to the extrapolations used in our experimental Tauc plots, is as high as 31 meV/GPa.

IV. CONCLUSIONS

In this work, we have investigated the effect of internal stresses on the electro-optical behaviour of Al-doped ZnO thin films, deposited by reactive magnetron sputtering to thicknesses from 300 to 500 nm. The following conclusions can be formulated:

- Growth-induced internal stresses in AZO films could be varied over a wide elastic range, from −600 MPa (compression) up to 800 MPa (tension), by appropriately choosing the oxygen partial pressure and the deposition temperature. Moreover, as a result of this tensile to compressive transition, a zero stress condition could be obtained for the first time in the case of AZO thin films.
- The resulting charge carrier mobilities, obtained independently from room temperature Hall measurements, were found to range between 5 and 25 $cm^2/V\,s$, the highest value occurring at the zero-stress condition. This correlation could be reproduced as well for undoped, pure ZnO thin films.
- Together with a charge carrier concentration of about $8 \times 1020\,cm^{-3}$, which was found to be temperature- rather than stress-activated, this gave rise to a resistivity of only 300 $\mu\Omega$·cm.
- From the stress-dependent optical transmission spectra in the range of 200–1000 nm, the pressure coefficient of the optical bandgap was estimated from the corresponding Tauc plots to be 31 meV/GPa, indicative for a very high strain-sensitivity as well.
- Al-doped ZnO can therefore be considered to be a highly stress-sensitive TCO material, which was attributed in

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turn to the piezo-electric nature of the wurtzite structure of the ZnO host crystal.

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[1]G. J. Exarhos and X.-D. Zhou, Thin Solid Films **515**, 7025 (2007).
[2]O. Isabella, J. Krc, and M. Zeman, Appl. Phys. Lett. **97**, 101106 (2010).
[3]N. L. Dehuff, E. S. Kettenring, D. Hong, H. Q. Chiang, J. F. Wager, R. L. Hoffman, C. H. Park, and D. A. Keszler, J. Appl. Phys. **97**, 064505 (2005).
[4]T. Minami, Semicond. Sci. Technol. **20**, S35 (2005).
[5]*Transparent Conductive Zinc Oxide: Basics and Applications in Thin Film Solar Cells*, edited by K. Ellmer, A. Klein, and B. Rech (Springer, Berlin, 2008).
[6]R. C. Powell, G. L. Dorer, U. Jayamaha, and J. J. Hanak, Final Technical Report No. NREL/SR-520-25422, National Renewable Energy Laboratory, Golden CO80401.
[7]E. Fortunato, D. Ginley, H. Hosono, and D. C. Paine, MRS Bull. **32**, 242 (2007).
[8]Y.-H. Hu, Y.-C. Chen, H.-J. Xu, H. Gao, W.-H. Jiang, F. Hu, and Y.-X. Wang, Engineering **2**, 973 (2010).
[9]S. Michotte and J. Proost, Sol. Energy Mater. Sol. Cells **98**, 253 (2012).
[10]F. Spaepen, Acta Mater. **48**, 31–42 (2000).
[11]C. V. Thompson, Ann. Rev. Mater. Sci. **30**, 159–190 (2000).
[12]R. Abermann, R. Koch, and R. Kramer, Thin Solid Films **58**, 365 (1979).
[13]M. Doerner and W. D. Nix, CRC Crit. Rev. Solid State Mater. Sci. **14**, 225 (1988).
[14]J. Proost and F. Spaepen, J. Appl. Phys. **91**, 204 (2002).
[15]B. D. Huey, D. Lisjak, and D. A. Bonnell, J. Am. Ceram. Soc. **82**, 1941 (1999).
[16]J. Hinze and K. Ellmer, J. Appl. Phys. **88**, 2443 (2000).
[17]R. J. Drese and M. Wuttig, J. Appl. Phys. **98**, 073514 (2005).
[18]A. Asadov, W. Gao, Z. Li, J. Lee, and M. Hodgson, Thin Solid Films **476**, 201 (2005).
[19]C. Guillén and J. Herrero, Thin Solid Films **515**, 640 (2006).
[20]A. I. Ievtushenko *et al.*, Thin Solid Films **518**, 4529 (2010).
[21]A. Walsh, J. L. F. Da Silva, and S.-H. Wei, J. Phys.: Condens. Matter **23**, 334210 (2011).
[22]K. Ellmer and R. Mientus, Thin Solid Films **516**, 4620 (2008); **516**, 5829 (2008).
[23]X. B. Zhang, Z. L. Pei, J. Gong, and C. Sun, J. Appl. Phys. **101**, 014910 (2007).
[24]J. A. Floro, S. J. Hearne, J. A. Hunter, P. Kotula, E. Chason, S. C. Seel, and C. V. Thompson, J. Appl. Phys. **89**, 04886 (2001).
[25]E. Chason and B. W. Sheldon, Surf. Eng. **19**, 387 (2003).
[26]K. Kamioka, T. Oga, Y. Izawa, K. Kuriyama, and K. Kushida, Nucl. Instrum. Methods Phys. Res. Sec., B **307**, 366 (2013).
[27]Y. Li, Q. Huang, and X. Bi, J. Mater. Sci.: Mater. Electron. **24**, 79 (2013).
[28]K. Ellmer, J. Phys. D: Appl. Phys. **33**, R17 (2000).
[29]E. Burstein, Phys. Rev. **93**, 632 (1954); T. S. Moss, Proc. Phys. Soc. London Ser. B **67**, 775 (1954).
[30]J. Tauc, Mater. Res. Bull. **5**, 721 (1970).
[31]J. Jia, A. Takasaki, N. Oka, and Y. Shigesato, J. Appl. Phys. **112**, 013718 (2012).
[32]S. Brehme, F. Fenske, W. Fuhs, E. Nebauer, M. Poschenrieder, B. Selle, and I. Sieber, Thin Solid Films **342**, 167–173 (1999).
[33]Z. Qiao, C. Agashe, and D. Mergel, "Dielectric modeling of transmittance spectra of thin ZnO:Al films," Thin Solid Films **496**, 520 (2006).
[34]*Numerical Data and Functional Relationships in Science and Technology*, Landoldt-Börnstein New Series, Vol. 17b, edited by H. Weiss, M. Schultz, and O. Madelung (Springer-Verlag, Berlin, 1982), p. 35.
[35]B. E. Sernelius, K.-F. Berggren, Z.-C. Jin, I. Hamberg, and C. G. Granqvist, Phys. Rev. B **37**, 10244 (1988).
[36]J. A. Sans, J. F. Sánchez-Royo, A. Segura, G. Tobias, and E. Canadell, Phys. Rev. B **79**, 195105 (2009).
[37]J. G. Lu, S. Fujita, T. Kawaharamura, H. Nishinaka, Y. Kamada, T. Ohshima, Z. Z. Ye, Y. J. Zeng, Y. Z. Zhang, L. P. Zhu, H. P. He, and B. H. Zhao, J. Appl. Phys. **101**, 083705 (2007).

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